MATH 350-2 Advanced Calculus

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Outline

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Integrals

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Most of these generalize the Riemann integral in some sense.

Partitions

Definition

A **partition** P of $[a, b]$ Is a set of points

$$\{x_0, x_1, \ldots, x_n\}$$

with

$$a = x_0 < x_1 < x_2 < \cdots < x_n = b.$$

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A set of **sample points** for P is a set $\{t_1, \ldots, t_n\}$ with $t_k \in [x_{k-1}, x_k]$ for all $1 \leq k \leq n$. A partition P' is a **refinement** of P if $P \subseteq P'$.

Riemann-Stieltjes sums

Notation: given a real-valued function α on $[a, b]$

$$\Delta\alpha_k(P) = \alpha(x_k) - \alpha(x_{k-1}), \quad k = 1, 2, \ldots, n.$$

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Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$.

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Definition

Let f and α be real-valued functions on $[a, b]$, and let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. A sum of the form

$$S(P, f, \alpha, \{t_k\}) = \sum_{k=1}^{n} f(t_k)\Delta\alpha_k(P),$$

for some set of sample points $\{t_k\}$ of P.

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Let f and α be real-valued functions on $[a, b]$. We say f is **Riemann integrable** with respect to α on $[a, b]$ if there exists $A \in \mathbb{R}$ with the following property: for all $\epsilon > 0$, there exists a partition P_ϵ of $[a, b]$ such that for all refinements P of P_ϵ

$$|S(P, f, \alpha, \{t_k\}) - A| < \epsilon,$$

for all sets of sample points $\{t_k\}$ of P.

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Notation: in this case A is denote by $\int_a^b f d\alpha$ or $\int_a^b f(x) d\alpha(x)$

Challenge!

Problem

Suppose that a, b are real numbers with $a < b$ and let α be a real-valued function on $[a, b]$.
Prove that a constant function $f(x) = c$ is integrable with respect to α on $[a, b]$ and

$$\int_a^b f d\alpha = c\alpha(b) - c\alpha(a).$$

Challenge!

Problem

Suppose that a, b are real numbers with $a < b$ and let α be a constant function on $a < b$.
If $f(x)$ is a function on $[a, b]$, show that $f(x)$ is integrable with respect to α on $[a, b]$.

$$\text{What is } \int_a^b f d\alpha?$$

Challenge!

Problem

Suppose that a, b, c are real numbers with $a < c < b$ and let α be the step function

$$\alpha(x) = \begin{cases} 0, & x < c \\ 1, & x \geq c \end{cases}$$

Show that if $f(x)$ is a real function on $[a, b]$ which is continuous at c, then f is integrable with respect to α on $[a, b]$.

$$\text{What is } \int_a^b f d\alpha?$$

Linearity of integration

Theorem

Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.

Linearity of integration

Theorem

Suppose that f and g are Riemann integrable with respect to α on $[a, b]$.
Then for any $c_1, c_2 \in \mathbb{R}$, $c_1 f + c_2 g$ is Riemann integrable with respect to α and

$$\int_a^b (c_1 f + c_2 g) d\alpha = c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha.$$

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Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.
Choose a partition P'_ϵ such that

$$|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| < \epsilon/2|c_1|,$$

for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P.

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for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P.
Also, choose a partition P''_ϵ such that

$$|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| < \epsilon/2|c_2|,$$

for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P.

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Let $P_\epsilon = P'_\epsilon \cup P''_\epsilon$.

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Proof.

Let $\epsilon > 0$ and WLOG assume $c_1, c_2 \neq 0$.
Choose a partition P'_ϵ such that

$$|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha| < \epsilon/2|c_1|,$$

for all refinements P of P'_ϵ and sets of sample points $\{t_k\}$ of P.
Also, choose a partition P''_ϵ such that

$$|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha| < \epsilon/2|c_2|,$$

for all refinements P of P''_ϵ and sets of sample points $\{t_k\}$ of P.
Let $P_\epsilon = P'_\epsilon \cup P''_\epsilon$. Then any refinement P of P_ϵ, is a refinement of P'_ϵ and P''_ϵ. □

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Therefore if P is a refinement of P_ϵ, and $\{t_k\}$ is a set of sample points of P, then by the triangle inequality

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Therefore if P is a refinement of P_ϵ, and $\{t_k\}$ is a set of sample points of P, then by the triangle inequality

$$
\begin{aligned}
&\left|S(P, c_1 f + c_2 g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)\right| \\
&= \left|c_1 S(P, f, \alpha, \{t_k\}) + c_2 S(P, g, \alpha, \{t_k\}) - (c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha)\right| \\
&= \left|c_1 S(P, f, \alpha, \{t_k\}) - c_1 \int_a^b f d\alpha + c_2 S(P, g, \alpha, \{t_k\}) - c_2 \int_a^b g d\alpha\right| \\
&\leq |c_1| \cdot \left|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha\right| + |c_2| \cdot \left|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha\right| \\
&< \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}
$$

Linearity of integration

Proof.

Therefore if P is a refinement of P_ϵ, and $\{t_k\}$ is a set of sample points of P, then by the triangle inequality

$$\begin{aligned}
&\left|S(P, c_1 f + c_2 g, \alpha, \{t_k\}) - \left(c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha\right)\right| \\
&= \left|c_1 S(P, f, \alpha, \{t_k\}) + c_2 S(P, g, \alpha, \{t_k\}) - \left(c_1 \int_a^b f d\alpha + c_2 \int_a^b g d\alpha\right)\right| \\
&= \left|c_1 S(P, f, \alpha, \{t_k\}) - c_1 \int_a^b f d\alpha + c_2 S(P, g, \alpha, \{t_k\}) - c_2 \int_a^b g d\alpha\right| \\
&\leq |c_1| \cdot \left|S(P, f, \alpha, \{t_k\}) - \int_a^b f d\alpha\right| + |c_2| \cdot \left|S(P, g, \alpha, \{t_k\}) - \int_a^b g d\alpha\right| \\
&< \epsilon/2 + \epsilon/2 = \epsilon.
\end{aligned}$$

Since $\epsilon > 0$ was arbitrary, this proves the theorem. $\square$

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Theorem

Suppose that f is Riemann integrable with respect to both α and β on $[a, b]$.

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Theorem

Suppose that f is Riemann integrable with respect to both α and β on $[a, b]$.
Then for any $c_1, c_2 \in \mathbb{R}$, f is Riemann integrable with respect to $c_1\alpha + c_2\beta$ and

$$\int_a^b f d(c_1\alpha + c_2\beta) = c_1 \int_a^b f d\alpha + c_2 \int_a^b f d\beta.$$

Challenge!

Problem

Prove the last theorem!

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$.

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ and that $c \in (a, b)$. If $\int_a^c f d\alpha$ and $\int_b^c f d\alpha$ exist, then $\int_a^c f d\alpha$ exists and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Linearity of integration

Theorem

Suppose that f is a real-valued function on $[a, b]$ *and that* $c \in (a, b)$. *If* $\int_a^c f d\alpha$ *and* $\int_b^c f d\alpha$ *exist, then* $\int_a^c f d\alpha$ *exists and*

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Conversely, if $\int_a^c f d\alpha$ *exists then* $\int_a^c f d\alpha$ *and* $\int_b^c f d\alpha$ *exist and*

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

Integration by parts

The integrand and integrator play dual roles:

Theorem

Integration by parts Let f be integrable with respect to α on $[a, b]$.
Then α is integrable with respect to f on $[a, b]$ and

$$\int_a^b f d\alpha = f(b)\alpha(b) - f(a)\alpha(a) - \int_a^b \alpha df.$$

Relation to Riemann integrals

Theorem

Let f be integrable with respect to α on $[a, b]$.
If α is continuous on $[a, b]$ and continuously differentiable on (a, b), then $f(x)\alpha'(x)$ is integrable with respect to x on $[a, b]$ and

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x)dx.$$